

FuseRAG: Parallel Multi-Source Knowledge Fusion for Hybrid Graph-Augmented Retrieval-Augmented Generation

1st Arrival Dwi Sentosa

*School of Electrical Engineering and Informatics
Bandung Institute of Technology
Bandung, Indonesia
arrivaldwi@itb.ac.id*

Abstract—Retrieval-Augmented Generation (RAG) grounds LLM responses in external knowledge, yet conventional vector-only retrieval lacks cross-document reasoning, and GraphRAG’s map-reduce community synthesis introduces $O(N)$ LLM calls per query. We present FuseRAG, a hybrid retrieval architecture that fuses knowledge graph-enriched vector search with hierarchical community detection via parallel multi-source context assembly into a single LLM call. In a controlled six-system benchmark where all architectures—conventional RAG, GraphRAG, HippoRAG, Self-RAG, Adaptive-RAG, and FuseRAG—share the same LLM (Gemini 2.5 Flash) and knowledge base, FuseRAG achieves 56% mean keyword relevance versus 36–47% for alternatives across five enterprise query scenarios, with 45.0 mean attribution signals per response versus ≤ 0.6 for all others. FuseRAG replaces GraphRAG’s per-query map-reduce with $O(1)$ PostgreSQL full-text search over pre-computed community summaries and supports incremental document updates in 25s without full corpus reindexing. We provide per-scenario breakdowns, component-level latency profiling, and cost projections, and we discuss the significant limitations of the evaluation, including small corpus size, keyword-based metrics, and prompt-level confounds that may inflate attribution scores.

Index Terms—retrieval-augmented generation, knowledge graph, community detection, enterprise search, graph-based retrieval, hybrid retrieval

I. INTRODUCTION

Large language models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation [1], [2]. However, they suffer from hallucination, knowledge staleness, and inability to access private enterprise data. Retrieval-Augmented Generation (RAG) [3] addresses these limitations by grounding LLM responses in retrieved documents, but the effectiveness of RAG depends fundamentally on retrieval quality.

Conventional RAG systems employ dense vector similarity search to identify the top- k most relevant text chunks for a given query. While effective for factual, single-document lookups, this approach has well-documented limitations: (1) inability to synthesize information across multiple documents, (2) lack of entity-relationship awareness, and (3) poor performance on thematic or global queries that require broad organizational understanding [4].

Microsoft Research introduced GraphRAG [4], which addresses these limitations by constructing a knowledge graph from source documents, applying the Leiden community detection algorithm [5] to identify thematic clusters, generating LLM-produced summaries for each community, and employing a map-reduce query pattern that sends each community summary to the LLM independently (map step) before synthesizing a final answer (reduce step). While this approach significantly improves answer quality for global and thematic queries, it introduces two critical drawbacks: (1) query latency scales linearly with the number of communities due to $O(N)$ LLM calls per query, and (2) the full knowledge graph must be rebuilt from scratch when source documents change, as no incremental update mechanism exists.

In this paper, we present FuseRAG, a hybrid retrieval architecture that achieves the cross-document reasoning capabilities of GraphRAG while maintaining competitive latency and supporting incremental updates. Our key contributions are:

- 1) A **parallel multi-source fusion** architecture that retrieves context from eight heterogeneous data sources concurrently (vector store, knowledge graph, community summaries, structured databases, activity logs, web search, academic search, and external databases) and assembles them into a single LLM generation call.
- 2) A **community global search** mechanism that replaces GraphRAG’s $O(N)$ map-reduce LLM pattern with $O(1)$ PostgreSQL full-text search over pre-computed community summaries, eliminating per-query LLM overhead for community-level reasoning.
- 3) An **incremental ingestion pipeline** with entity deduplication and community rebuilding that processes document updates in seconds without full corpus reindexing.
- 4) A comprehensive **empirical benchmark** comparing six retrieval architectures (conventional RAG, GraphRAG, HippoRAG, Self-RAG, Adaptive-RAG, and FuseRAG) using the same LLM, same knowledge base, and same evaluation methodology across five retrieval scenarios.